EMS611U: Conceptual Design of Aircraft

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Week 8 – Part 2

Thrust-to-Weight Ratio & Wing Loading

Wing Loading

Where We Are: Road Map

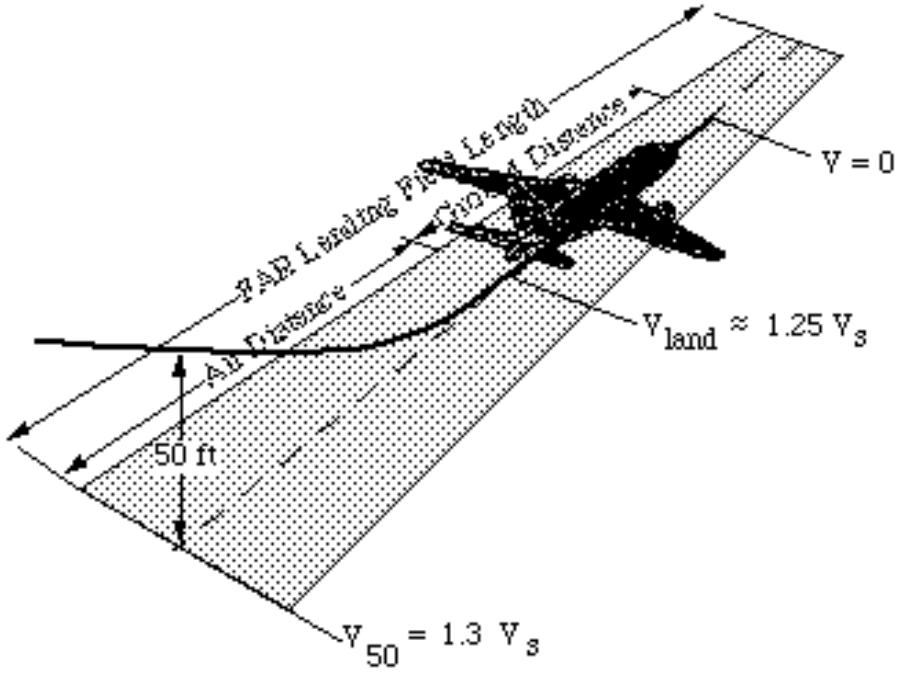
Stall speed ✓	Take-off distance ✓	Catapult take-off ✓
Landing distance	Arrested landing	Cruise (best range)
Loiter (endurance)	Turning (instantaneous, sustained)	Climb, glide and ceiling

Select the **lowest** W/S (all converted to take-off conditions), then **recheck T/W**

Wing Loading: Landing

Landing Distance: Definitions

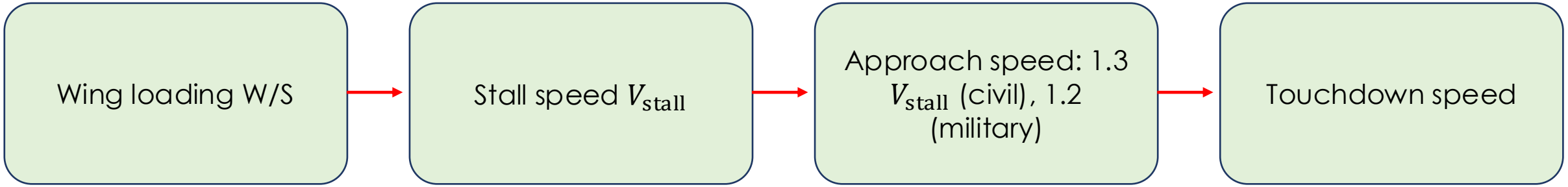
- **Landing ground roll:** touchdown to a complete stop.
- **FAR 23 landing field length:** from a **50-ft obstacle** on a **3°** glide path; flare, touch down at about **1.15** V_{stall}. Obstacle clearance roughly **doubles** the ground roll.
- **FAR 25:** the same, with **two-thirds** added as a safety margin.
- **Military:** set in the request for proposals; usually like FAR 23.



FAR landing field length

Wing Loading: Landing

Landing Distance Depends on the Touchdown Speed



- Kinetic energy $\frac{1}{2}mV^2$ must be removed by the brakes (and reversers) with a roughly constant force, so the distance varies as the **square** of the touchdown speed.

Double the touchdown speed → four times the runway.

Wing Loading: Landing

First Estimate of the Landing Distance

- A reasonable first guess of the total landing distance, including obstacle clearance (Raymer, from statistics):

$$S_{\text{landing}} \approx 0.3\, V_{\text{approach}}^2 \quad [\text{ft}, V \text{ in knots}]$$

- Approximately true for FAR 23 and military aircraft without reversers, and for FAR 25 aircraft with thrust reversers (the extra two-thirds is cancelled by the reversers).

Illustration (my numbers): V_{approach} = 130 kt → 0.3 × 130² ≈ **5,070 ft** (1,545 m)

Wing Loading: Landing

A Better Estimate:

$$S_{\text{landing}} = 80\left(\frac{W}{S}\right)\left(\frac{1}{\sigma C_{L_{\max}}}\right) + S_a$$

1st term: ground roll: absorbing the kinetic energy at touchdown speed

2nd term: air distance from the 50-ft obstacle to touchdown

Approach type	S_a [ft]
Airliner, 3° glide slope	1000
General aviation, power-off	600
STOL, 7° glide slope	450

Wing Loading: Landing

Landing Distance: Units, Reversers and FAR 25

- Units: S in ft, W/S in lb/ft². σ = density ratio at the landing field.
- With thrust reversers or reversible-pitch propellers: multiply the **ground term** (1st term) by **0.66**.
- **Commercial (FAR 25):** multiply the total distance by **1.67**.

Check on S_a: $S_a \approx 50 \text{ ft} / \tan\gamma$

3° glide slope → 954 ft ≈ 1000 ft

7° glide slope → 407 ft ≈ 450 ft

Wing Loading: Landing

From Required Landing Distance to Wing Loading

Step 1. Solve Eq. 5.11 for W/S (first remove the ×1.67 or ×0.66 factors, if they apply):

$$\left(\frac{W}{S}\right)_{\text{land}} = \frac{\sigma C_{L_{\max}} \left(S_{\text{landing}} - S_a\right)}{80}$$

Step 2. Convert to take-off conditions: divide by the ratio of landing to take-off weight:

$$\left(\frac{W}{S}\right)_{TO} = \frac{(W/S)_{\text{land}}}{W_{\text{land}}/W_{TO}}$$

- ≈ 1.0: propeller aircraft, jet trainers
- ≈ 0.85: most jets
- military: full payload, ~50% fuel

Wing Loading: Landing

Arrested Landing (Carrier-based Aircraft)

- A hook catches a cable that is paid out against a damper: the aircraft stops in a very short distance.
- Approach speed ($1.15\,V_{\text{stall}}$) **is** the touchdown speed: no flare, so there is speed to go around if the hook misses.
- The gear limits the landing weight : weight → engagement speed → ÷ 1.15 → stall speed.

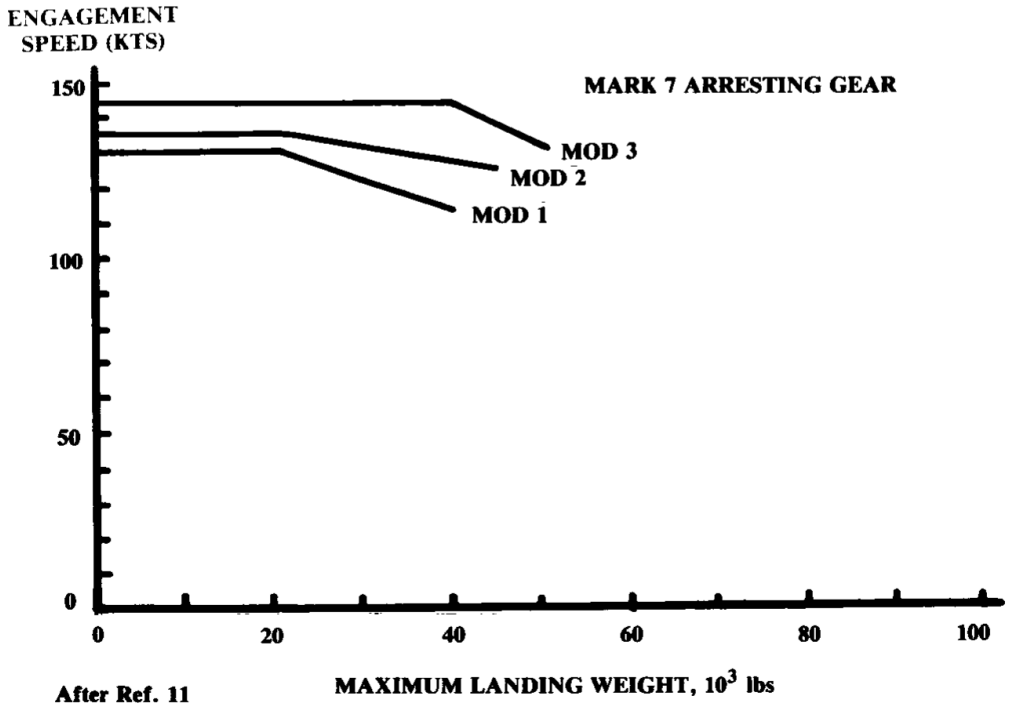


Fig. 5.6 Arresting gear weight limits (Raymer)

$$\left(\frac{W}{S}\right)_{\text{landing}} = \frac{1}{2}\rho\left(\frac{V_{\text{engage}}}{1.15}\right)^2 C_{L_{\max}}$$

Then divide by $W_{\text{land}}/W_{\text{TO}}$ for take-off conditions.

Wing Loading: Cruise and Loiter

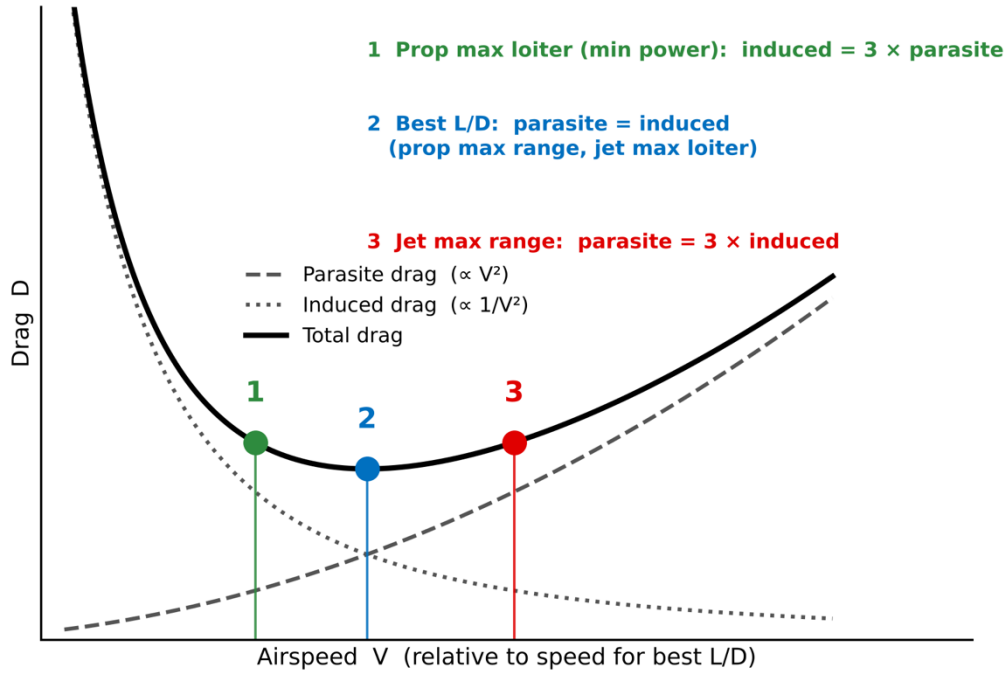
Wing Loading for Cruise: the Drag Build-up

- To maximise range, choose W/S so that the aircraft cruises at a high L/D.

- Drag = **parasite** + **induced**
  - Parasite drag ∝ V²: friction and pressure drag grow with speed.
  - Induced drag ∝ 1/V²: slow flight needs a high C_L, which is costly (tip vortices).

$$D = qSC_{D_0} + qS\frac{C_L^2}{\pi Ae}$$

- Best L/D at the speed where the two are **equal.**



Drag against speed: where each optimum sits

Wing Loading: Cruise and Loiter

First Estimates: C_{D0} and Oswald Efficiency e

Aircraft	C_{D0}
Clean jet	0.015
Clean propeller aircraft	0.02
Propeller, fixed landing gear	0.03

Aircraft	e
Fighter	0.6
Other aircraft	0.8

- Aspect ratio A: a long, thin wing (high A) acts on a larger mass of air and needs less downwash: **less induced drag**.

Wing Loading: Cruise and Loiter

Maximum Range: Propeller Aircraft (Equations 5.12 and 5.13)

- **Propeller aircraft** lose efficiency as speed rises, so they get the best range at **best L/D**: parasite drag = induced drag

$$qSC_{D_0} = qS\frac{C_L^2}{\pi Ae} \qquad (5.12)$$

- In cruise L = W, so C_L = (W/S)/q. Substituting into 5.12:

$$\frac{W}{S} = q\sqrt{\pi Ae C_{D_0}} \qquad (5.13)$$

q in psf → W/S in lb/ft²; q in Pa → N/m²

Wing Loading: Cruise and Loiter

Maximum Range: Jet Aircraft (Equation 5.14)

- A jet burns fuel in proportion to thrust, so a little more speed pays off (range ∝ V × L/D).
- The optimum is where parasite drag is **three times** the induced drag:

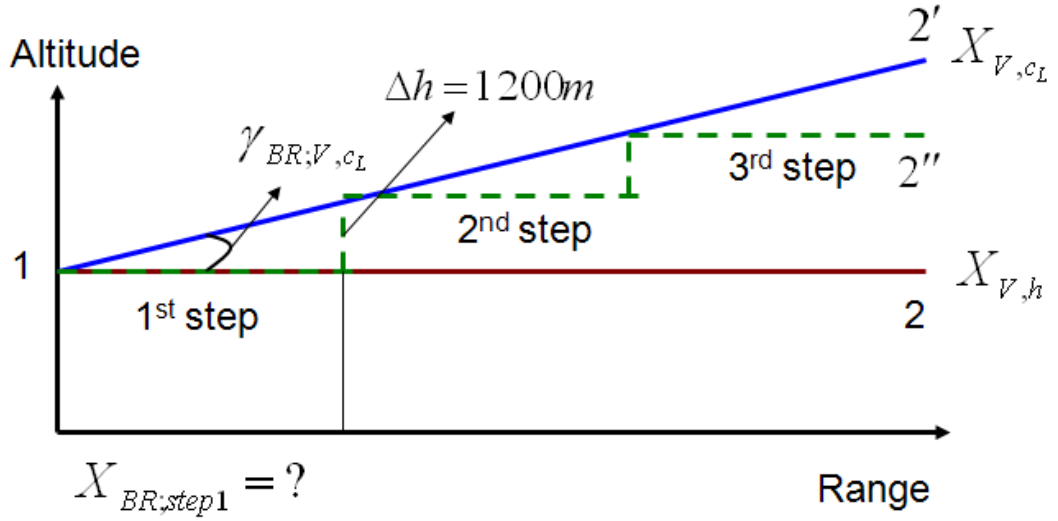
$$\frac{W}{S} = q\sqrt{\frac{\pi A e C_{D_0}}{3}} \qquad (5.14)$$

Illustration (my numbers): compared with best L/D, the speed is **1.32×** (3^¼) and L/D is **0.87×**: 13% less L/D for 32% more speed.

Wing Loading: Cruise and Loiter

Cruise-climb and Stair-step Climb

- Fuel is burned, weight falls: the wing loading falls during cruise.
- To stay at the best C_L, q must fall too (W/S = q C_L): **climb to thinner air**. This is the **cruise-climb**.
- Air-traffic control wants set altitudes: the pilot requests steps: the **stair-step climb**.



Cruise-climb (blue) and stair-step climb (green)

Wing Loading: Cruise and Loiter

Wing Loading for Loiter Endurance (Jet Aircraft)

- Typical requirement: about 20 min of loiter before landing. Unless loiter is a large part of the mission, optimise for cruise.
- Patrol (ASW), airborne command posts and intelligence aircraft care more about time on station.
- **Jet:** fuel flow follows thrust (≈ drag): minimise drag, i.e. best L/D (parasite = induced).

$$\frac{W}{S} = q\sqrt{\pi A e C_{D_0}} \qquad (5.15)$$

same as maximum range for a propeller aircraft (5.13)

Wing Loading: Cruise and Loiter

Wing Loading for Loiter Endurance (Propeller Aircraft)

- **Propeller:** fuel flow follows power (≈ drag × speed): minimise power by flying slower, where induced drag = **3 × parasite** drag.

$$\frac{W}{S} = q\sqrt{3\pi A e C_{D_0}} \qquad (5.16)$$

also the wing loading for minimum power required

Wing Loading: Cruise and Loiter

Cruise and Loiter: Summary and Assumptions

Mission	Aircraft	Optimum condition	W/S
Range	Propeller	parasite = induced	$q\sqrt{(\pi A e C_{D0})}$
Range	Jet	parasite = 3 × induced	$q\sqrt{(\pi A e C_{D0}/3)}$
Loiter	Jet	parasite = induced	$q\sqrt{(\pi A e C_{D0})}$
Loiter	Propeller	induced = 3 × parasite	$q\sqrt{(3\pi A e C_{D0})}$

- Loiter speed ≈ **150–200 kt** (turboprops, jets); **80–120 kt** (piston props).
- Altitude: jets 30,000–40,000 ft; turbocharged piston: its limit; non-turbocharged: sea level.
- Convert to take-off conditions: divide by W_{avg}/W_{TO} ≈ **0.85**.
- Loiter alone rarely sizes the wing: it is usually a "fallout" of the cruise choice.

Wing Loading: Cruise and Loiter

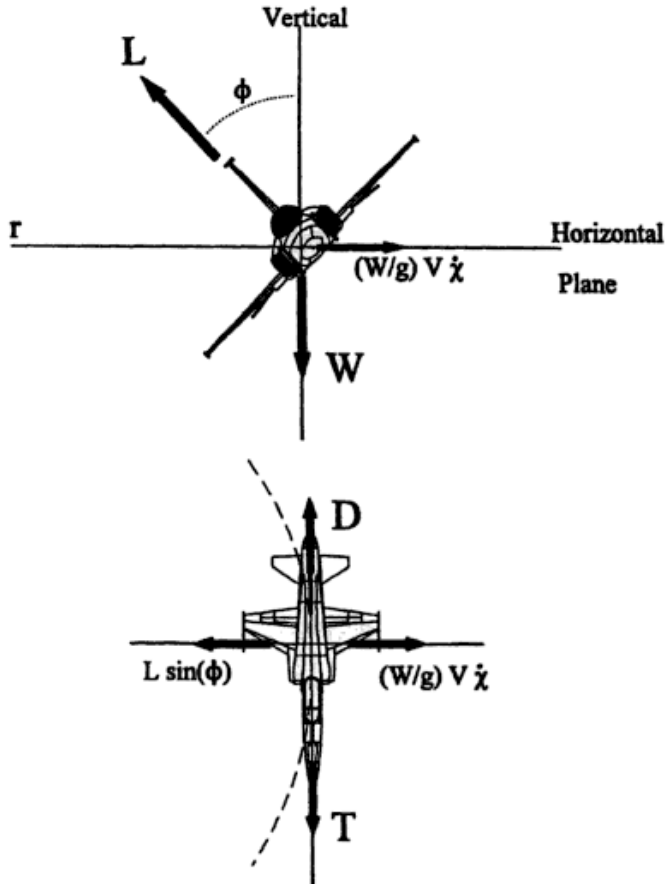
Worked Example: Jet Range, and a Caution

Illustration (my numbers): jet, A = 9, e = 0.8, C_{D_0} = 0.015, M = 0.8 at 35,000 ft (q ≈ 223 psf)
W/S = 223 √(π ·9 ·0.8 ·0.015 / 3) ≈ **75 lb/ft²** (366 kg/m²)
a propeller formula would give ≈ 130 lb/ft² (√3 higher)

Caution: Equations 5.13–5.16, 5.22 and 5.32 optimise only a **part of the mission**: if they give W/S far below Table 5.5, they may be ignored.

Wing Loading: Turning

Turn Rate: Forces in a Level Turn



Forces in a level turn

- Turn rate $\dot{\phi}$ decides a dogfight between evenly matched aircraft: an advantage of **2°/s** is significant.
- Level turn: lift does two jobs. $\boldsymbol{L\cos\phi = W}$ holds the aircraft up; the rest pulls it sideways: **g √(n²−1)**, with load factor $n = L/W$.

Illustration (my numbers): n = 2 → bank 60°, sideways acceleration 1.73 g

n = 5.8 → bank 80°, sideways acceleration 5.7 g

Wing Loading: Turning

Turn Rate and Turn Radius: Equations 5.17 and 5.18

- Turn rate = radial acceleration ÷ speed (rad/s; × 57.3 for deg/s):

$$\dot{\phi} = \frac{g\sqrt{n^2 - 1}}{V} \qquad (5.17)$$

$$R = \frac{V^2}{g\sqrt{n^2 - 1}}$$

turn radius

- The wing supplies the load factor:

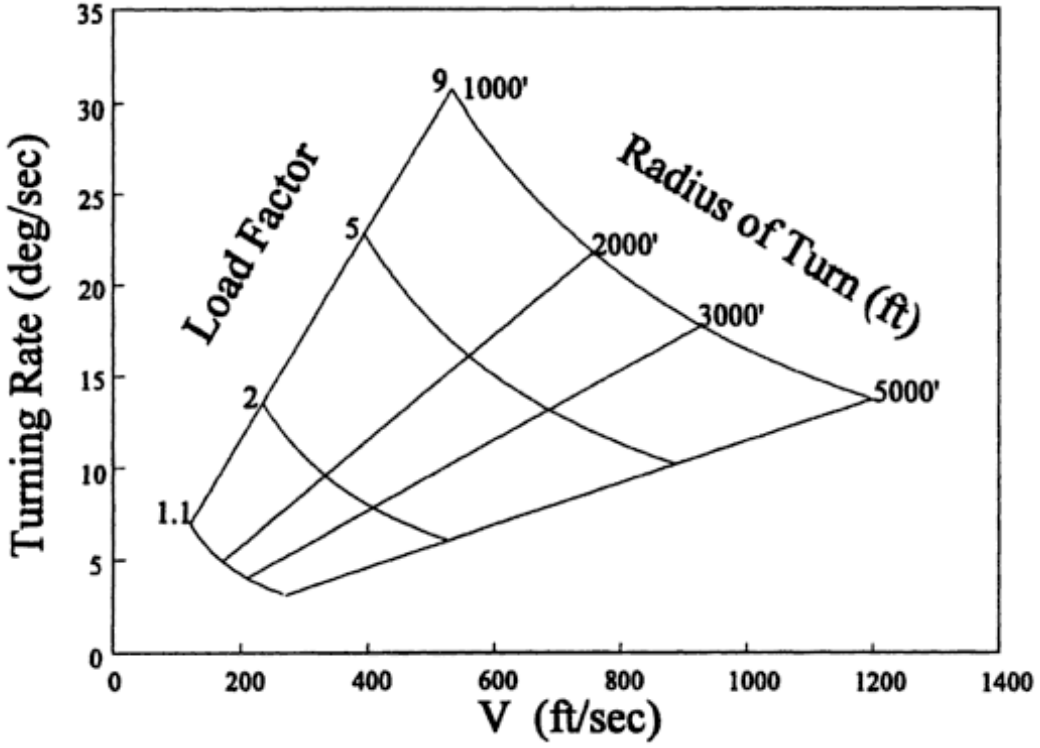
$$n = \frac{qC_L}{W/S} \qquad (5.18)$$

Illustration (my numbers): n = 2, V = 400 ft/s: $\dot{\phi}$ = 1.73 × 32.2 / 400 = 0.139 rad/s = **8°/s**; R ≈ 2,870 ft

Wing Loading: Turning

Instantaneous and Sustained Turn Rates: Trends

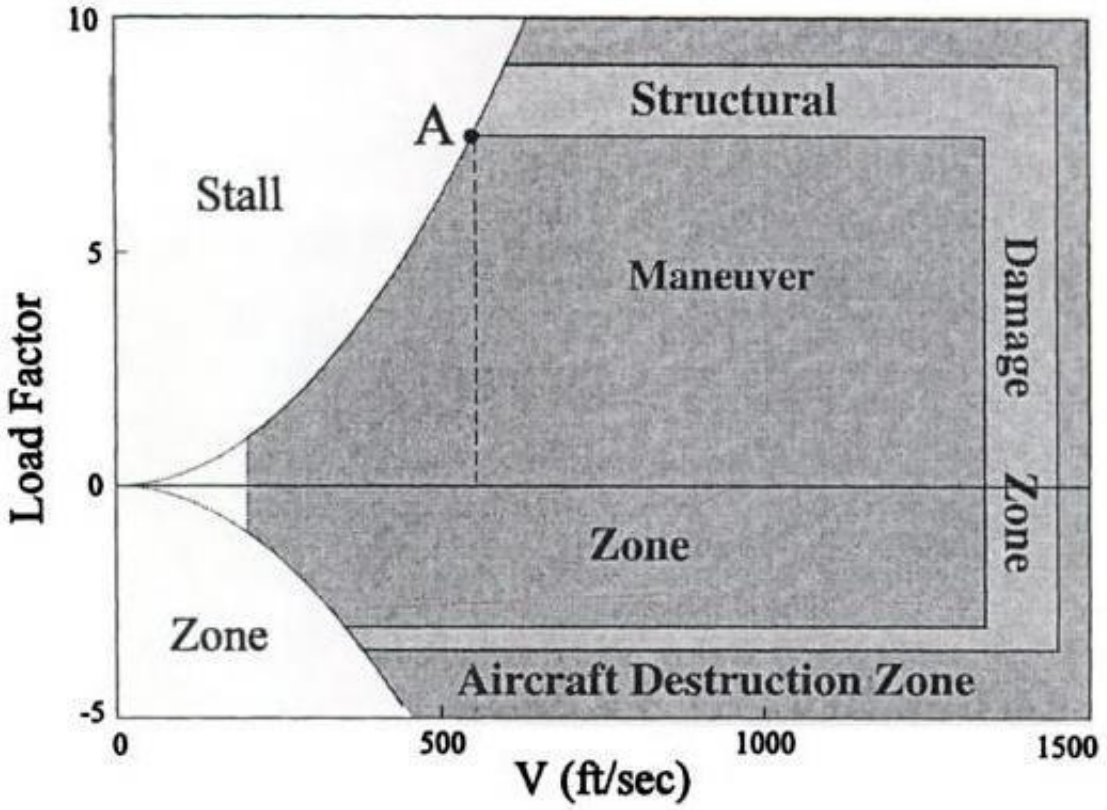
- **Sustained turn:** thrust balances drag: speed and altitude are held.
- **Instantaneous turn:** the highest rate possible (lift and structure limits): speed or altitude is lost.
- Higher n or lower V → **faster, tighter** turn
  - $\dot{\phi} = \frac{g\sqrt{n^2-1}}{V}$ & $R = \frac{V^2}{g\sqrt{n^2-1}}$
- Best turn: the highest n at the lowest V that allows it: the corner.



Turn rate against speed, with load-factor and radius lines

Wing Loading: Turning

Load Factor Limits and Corner Speed



V–n diagram; A = corner point

- Instantaneous turn rate is limited only by maximum lift, up to the speed where lift exceeds what the structure can carry.
- A fighter is designed for **7.33 g** (newer ones 8–9 g) at combat weight.
- **Corner speed:** maximum lift just equals the allowed load factor: the **best turn rate**.

$$n_{\text{lift limit}} = \frac{qC_{L_{\max}}}{W/S} \propto V^2$$

Wing Loading: Turning

Wing Loading for a Required Instantaneous Turn

$$\dot{\phi} = \frac{g\sqrt{n^2 - 1}}{V} \qquad (5.17)$$

- Solve Eq. 5.17 for the load factor at the required turn rate and speed:

$$n = \sqrt{\left(\frac{\dot{\phi}\, V}{g}\right)^2 + 1} \qquad (5.19)$$

- If n exceeds the ultimate load factor of the design requirements, somebody has made a mistake. Then, from Eq. 5.18 with $C_{L_{\max}}$:

$$n = \frac{qC_L}{W/S} \qquad (5.18)$$

$$\frac{W}{S} = \frac{qC_{L_{\max}}}{n} \qquad (5.20)$$

Wing Loading: Turning

Instantaneous Turn: $C_{L_{\max}}$ at Combat Conditions

- The only unknown is $C_{L_{\max}}$ at combat conditions (not the landing value): full flaps usually cannot be used; Mach effects and buffet limit it.
- Initial design values: **0.6–0.8** (simple trailing-edge flap); **1.0–1.5** (leading- and trailing-edge flap system).
- Divide by the combat-to-take-off weight ratio, **≈ 0.85** (external tanks dropped, 50% internal fuel gone). The result is the maximum W/S for the required turn.

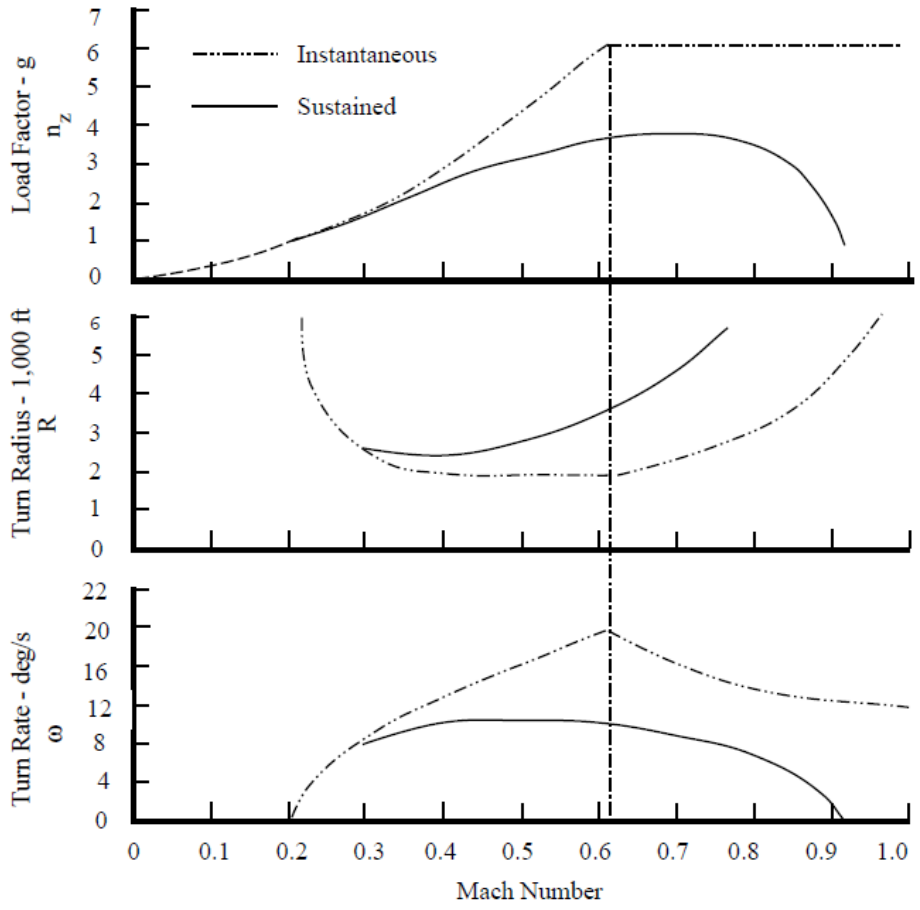
Wing Loading: Turning

Sustained Turn: Thrust Must Equal Drag

- Two aircraft pass in opposite directions: a 180° turn back takes about **10 s**. The one that slows below corner speed is at a disadvantage.
- Requirement: the maximum load factor held without slowing or losing altitude, e.g. **4–5 g at M 0.9, 30,000 ft**.
- Thrust = drag and lift = n × weight, so:

$$n = \left(\frac{T}{W}\right)\left(\frac{L}{D}\right) \qquad (5.21)$$

- Maximise T/W and L/D (best L/D: induced = parasite drag, Eq. 5.12).



Instantaneous and sustained: n, radius, rate

Wing Loading: Turning

Sustained Turn: Drag Balance

- With C_L = n (W/S)/q, thrust = drag: T = D = qSC_D0 + qS $C_L^2/(\pi Ae)$ (5.23). Per unit weight: **T/W = parasite term + induced term**:

$$\frac{T}{W} = \frac{qC_{D_0}}{W/S} + \frac{W}{S}\left(\frac{n^2}{q\pi Ae}\right) \qquad (5.24)$$

- 1st term falls with W/S: a small wing has less area to drag.
- 2nd term rises with W/S and n²: a small wing works harder to make n·W of lift.

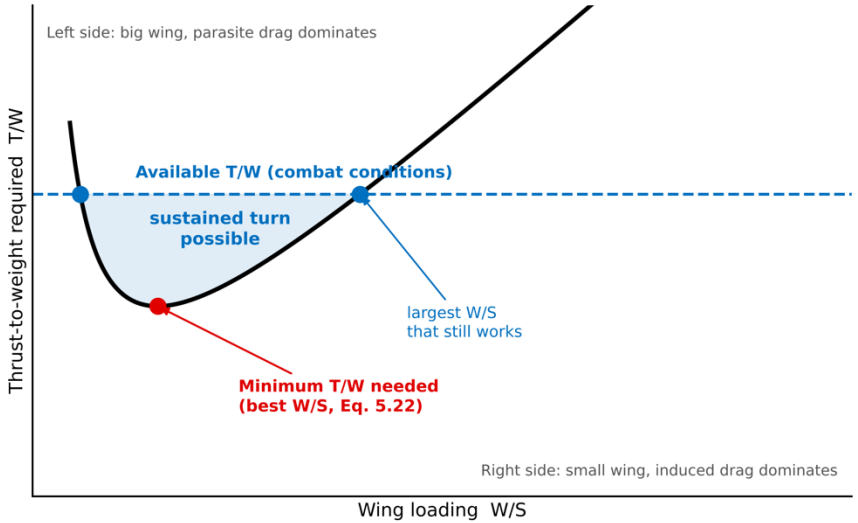


Illustration: T/W required for n = 5 (my numbers)

The sum is **U-shaped**: there is a W/S with the least T/W needed.

Wing Loading: Turning

Sustained Turn: Wing Loading for the Least Thrust

- Differentiating Eq. 5.24 with respect to W/S and setting it to zero gives the wing loading with the **smallest T/W** needed:

$$\frac{W}{S} = \frac{q}{n}\sqrt{\pi A e C_{D_0}} \qquad (5.22)$$

For n = 1 this is Eq. 5.13: the wing loading for best L/D in level flight.

Wing Loading: Turning

Sustained Turn: Using All the Available Thrust

- Eq. 5.22 gives the least thrust needed, regardless of the thrust available. To use all the thrust, solve Eq. 5.24 for W/S:

$$\frac{T}{W} = \frac{qC_{D_0}}{W/S} + \frac{W}{S}\left(\frac{n^2}{q\pi Ae}\right) \qquad (5.24)$$

$$\frac{W}{S} = \frac{\left(\frac{T}{W}\right) \pm \sqrt{\left(\frac{T}{W}\right)^2 - \frac{4n^2 C_{D_0}}{\pi Ae}}}{\frac{2n^2}{q\pi Ae}} \qquad (5.25)$$

- T/W is at combat conditions: divide the take-off T/W by the weight ratio and multiply by the thrust ratio.
- No solution if the root is negative: $\frac{T}{W} \geq 2n\sqrt{C_{D_0}/\pi Ae}$ (5.26).

Wing Loading: Turning

Sustained Turn: Which Root, and How Good is e?

- Use the **+ root** (largest W/S); the − root is the very large-wing solution.

Caution: e falls with C_L (flow separation): by **30% or more** at high angles of attack. The results are very sensitive to e: if W/S is far from historical values, e is probably unrealistic.

Wing Loading: Climb, Glide and Ceiling

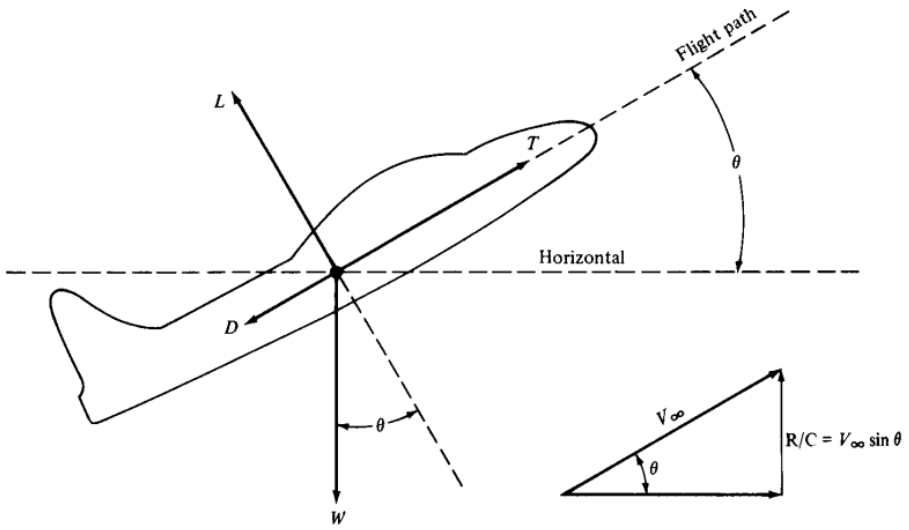
Climb and Glide: the Climb Gradient

- Rate of climb is a vertical speed (ft/min → ft/s). The climb gradient G = (R/C) ÷ V.
- The gradient is the **excess thrust divided by weight**:

$$G = \frac{T - D}{W} \quad (5.27)$$

$$\frac{D}{W} = \frac{T}{W} - G \quad (5.28)$$

$$\frac{D}{W} = \frac{qC_{D_0}}{W/S} + \frac{W}{S}\frac{1}{q\pi Ae} \quad (5.29)$$



Climb geometry

Wing Loading: Climb, Glide and Ceiling

Climb: Wing Loading for a Required Gradient (Equation 5.30)

- Equate 5.28 and 5.29 and solve for W/S:

$$\frac{W}{S} = \frac{\left[\left(\frac{T}{W}\right) - G\right] \pm \sqrt{\left[\left(\frac{T}{W}\right) - G\right]^2 - \frac{4C_{D_0}}{\pi Ae}}}{\frac{2}{q\pi Ae}} \qquad (5.30)$$

Same as Eq. 5.25 with n = 1 and T/W → (T/W) − G: the thrust left after the climb is paid for. T/W is at the flight condition: convert the resulting W/S to take-off weight.

Wing Loading: Climb, Glide and Ceiling

Climb: Minimum T/W

$$\frac{T}{W} \geq G + 2\sqrt{\frac{C_{D_0}}{\pi A e}} \qquad (5.31)$$

- No matter how clean the design, **T/W must exceed the climb gradient**.
- A very clean aircraft cruising fast on very low T/W climbs poorly: a 200-mph aeroplane on 20 hp will not climb like one that needs 200 hp (unless the latter weighs ten times more).

Wing Loading: Climb, Glide and Ceiling

Climb: Drag Increments, One Engine Out, Glide

Configuration	ΔC_{D0}	e
Take-off flaps	+0.02	−5%
Landing flaps	+0.07	−10%
Gear down	+0.02	—

- One engine out: a 3-engine aircraft keeps 2/3 of its T/W. Ignore windmilling-engine drag for now.
- **Glide:** set T/W = 0 and use a negative G (sink rate ÷ forward speed, same units).

Wing Loading: Climb, Glide and Ceiling

Maximum Ceiling

$$\frac{W}{S} = \frac{\left[\left(\frac{T}{W}\right) - G\right] \pm \sqrt{\left[\left(\frac{T}{W}\right) - G\right]^2 - \frac{4C_{D_0}}{\pi Ae}}}{\frac{2}{q\pi Ae}} \qquad (5.30)$$

- Use Eq. 5.30 with the T/W available at that altitude and G = 0 (level flight at the ceiling).
- A small climb rate is often required at the ceiling, e.g. **100 ft/min**: convert to G = (100/60 ft/s) ÷ V and include it in Eq. 5.30.
- High-altitude aircraft: the very low dynamic pressure may set the minimum W/S: at 100,000 ft and Mach 0.8, q ≈ **10 psf** (Raymer prints "psi").

Wing Loading: Climb, Glide and Ceiling

Ceiling: Wing Loading at Very High Altitude

$$\frac{W}{S} = q\sqrt{\pi A e C_{D_0}} \qquad (5.32)$$

$$\frac{W}{S} = qC_L \qquad (5.33)$$

- 5.32 may give a W/S so low as to be impractical: compare it with 5.33.
- Efficient high-altitude cruise: C_L near the airfoil design C_L: **0.5** typically; high-lift airfoils **1.0–1.4**.

Illustration: q = 10 psf, C_L = 0.5 → W/S = **5 lb/ft²** (24 kg/m²): a glider-like wing; with C_L = 1.2: 12 lb/ft²

Selection of T/W and W/S

Selecting the Design Values: Procedure

- ✓ **1.** Estimate T/W first (statistics, thrust matching).
- ✓ **2.** Find the maximum W/S each requirement allows: stall, take-off, landing, cruise, loiter, turn, climb, ceiling.
- ✓ **3. Convert every W/S to take-off conditions** before comparing.
- ✓ **4.** Select the **lowest**: the wing must be large enough for all flight conditions.
- ✓ **5.** A low W/S always raises weight and cost: if one requirement drives it, change that design assumption (e.g. a better high-lift system).
- ✓ **6.** Eqs. 5.13–5.16, 5.22, 5.32 optimise only part of the mission: ignore them if far below Table 5.5.
- ✓ **7. Recheck T/W** with the selected W/S: recalculate every equation that uses T/W. Then start initial sizing.

Selection of T/W and W/S

Summary: What Each Requirement Gives

Requirement	Eq.	Governing relation	Result
Stall speed	5.6	$W/S = \tfrac{1}{2}\rho V_{stall}^2 C_{Lmax}$	maximum W/S
Take-off	5.8 / 5.9	$W/S = (TOP)\,\sigma\, C_{LTO}\,(T/W)$	maximum W/S (given T/W)
Catapult	5.10	$W/S = \tfrac{1}{2}\rho(V_{end} + V_{wod})^2 C_{Lmax}/1.21$	maximum W/S
Landing	5.11	$S = 80\,(W/S)/(\sigma C_{Lmax}) + S_a$	maximum W/S
Arrested landing	5.6, Fig. 5.6	$W/S = \tfrac{1}{2}\rho(V_{engage}/1.15)^2 C_{Lmax}$	maximum W/S
Cruise (range)	5.13 / 5.14	$W/S = q\sqrt{(\pi A e C_{D0})}$ (÷√3 for jets)	optimum W/S
Loiter (endurance)	5.15 / 5.16	$W/S = q\sqrt{(\pi A e C_{D0})}$ (×√3 for props)	optimum W/S
Instantaneous turn	5.20	$W/S = qC_{Lmax}/n$	maximum W/S
Sustained turn	5.22, 5.25	$W/S = (q/n)\sqrt{(\pi A e C_{D0})}$, or the + root of 5.25	maximum W/S
Climb / glide	5.30	as 5.25 with n = 1 and T/W → T/W − G	maximum W/S
Ceiling	5.30, 5.32, 5.33	G = 0; $W/S = qC_L$	W/S at altitude

All results are converted to take-off conditions; then the lowest W/S is selected and T/W rechecked.

Selection of T/W and W/S

Key Take-aways

- ✓ T/W and W/S are the two numbers that set performance; they must be chosen together.
- ✓ **T/W** is defined at sea-level static take-off conditions: statistics and thrust matching give a first value; climb, take-off and turning check it.
- ✓ **W/S**: every requirement gives a limit. Convert all to take-off conditions and select the **lowest**.
- ✓ If one requirement forces an unreasonable W/S, change the design (high-lift devices, more thrust), not just the wing.
- ✓ **Recheck T/W** with the selected W/S, compare with real aircraft, then go on to initial sizing.